Number Logic

Year 4 Pt 2 · **Mark Scheme**

Lesson Worksheet — Answers

Purple — Accessible

1. 32 is the odd one out — the other four (15, 20, 25, 30) are all multiples of 5, but 32 is not.

2. 15 (the rule is: add the two numbers together).

3. 11 years old (Beth = 4×2 = 8, so Amir = 8+3 = 11).

4. C) 15 (the rule is: add the two numbers together).

5. 21 is the odd one out — the other four (12, 16, 18, 24) are all even numbers, but 21 is not.

6. 15 points (9+6 = 15).

7. 10 (same rule as rows 1 and 2: add the first two numbers to get the third).

8. 17 (the rule is: add the two numbers together).

9. A, B and D are correct — Priya > Sam > Tom, so Priya is older than Tom (A), Tom is the youngest (B), and Priya is the oldest of the three (D). C and E reverse true comparisons and are false.

10. D) 45 — the other four are all multiples of 10.

11. 15 marbles (5×3 = 15).

12. 16 (the rule is: add the two numbers together).

13. C) 5 (Lily = 7+2 = 9; Noah = 9-4 = 5).

14. 17 is the odd one out — the other four (9, 12, 15, 21) are all multiples of 3, but 17 is not.

15. C) 19 (the rule is: add the two numbers together).

Green — Standard

16. 40 is the odd one out — the other four (16, 25, 36, 49) are all square numbers, but 40 is not.

17. 8 (the rule is: subtract the smaller number from the larger).

18. 26 stickers (Zara = 8+5 = 13; Omar = 13×2 = 26).

19. C) 13 (the rule is: double the first number, then add the second).

20. 22 is the odd one out — the other four (12, 16, 20, 28) are all multiples of 4, but 22 is not.

21. 41 marks (Dev = 72+10 = 82; Asha = 82÷2 = 41).

22. 13 (same rule as rows 1 and 2: double the first number, then add the second).

23. 15 (the rule is: double the first number, then add the second).

24. A, C, E are correct (each gives 8 using the rule: subtract the smaller number from the larger).

25. D) 29 — the other four are all numbers whose digits add up to a multiple of 3.

26. 39 cakes (Table B = 18×3 = 54; Table C = 54-15 = 39).

27. 9 (the rule is: subtract the smaller number from the larger).

28. C) 15 (Max = 10+5 = 15; Ivy's age, 8, is not needed).

29. 21 is the odd one out — the other four (11, 13, 17, 19) are all prime numbers, but 21 is not.

30. C) 17 (the rule is: subtract the smaller number from the larger).

Red — Challenge

31. 24 is the odd one out — the other four (10, 15, 21, 28) are all triangular numbers (the sum of the first n whole numbers), but 24 is not.

32. 14 (the rule is: multiply the two numbers together, then subtract their sum from the result).

33. £75 (Class B = £45+15 = £60; Class C = £60÷2 = £30; A+C = £45+£30 = £75).

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34. C) 11 (the rule is: multiply the two numbers together, then subtract their sum from the result).

35. 27 is the odd one out — the other four (16, 25, 49, 81) are all square numbers, but 27 is not.

36. 48 years old (Leo = 9+3 = 12; Grandma = 12×4 = 48).

37. 16 (same rule as rows 1 and 2: square the first number, then subtract the second).

38. 30 (the rule is: add the two numbers together, then double the result).

39. A, B and E are correct — Omar > Priya = Sana > Tariq, so Omar is older than Sana (A), Priya is older than Tariq (B), and Omar is the oldest of all four (E). C and D reverse true comparisons and are false.

40. D) 51 — the other four are all prime numbers.

41. 19 (Bella = 6×2 = 12; Chen = 12-5 = 7; 12+7 = 19).

42. 26 (the rule is: square the first number, then subtract the second).

43. C) 30 (Red = 7×3 = 21; Green = 21+9 = 30).

44. 52 is the odd one out — the other four (18, 27, 36, 45) are all numbers whose digits add up to 9, but 52 is not.

45. C) 18 (the rule is: add the two numbers together, then double the result).

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Homework Worksheet — Answers

Purple — Accessible

1. 25 is the odd one out — the other four (14, 18, 22, 30) are all even numbers, but 25 is not.

2. 12 (the rule is: add the two numbers together).

3. 10 pencils (6+4 = 10).

4. C) 15 (the rule is: add the two numbers together).

5. 38 is the odd one out — the other four (25, 30, 35, 45) are all multiples of 5, but 38 is not.

6. 3 kg (7-4 = 3).

7. 11 (same rule as rows 1 and 2: add the first two numbers to get the third).

8. 12 (the rule is: add the two numbers together).

9. A, C, E are correct (each gives 14 using the rule: add the two numbers together).

10. E) 42 — the other four are all multiples of 10.

11. 17 pages (12+5 = 17).

12. 16 (the rule is: add the two numbers together).

13. C) 16 (8×2 = 16).

14. 20 is the odd one out — the other four (6, 9, 15, 24) are all multiples of 3, but 20 is not.

15. C) 19 (the rule is: add the two numbers together).

Green — Standard

16. 40 is the odd one out — the other four (9, 16, 36, 49) are all square numbers, but 40 is not.

17. 11 (the rule is: subtract the smaller number from the larger).

18. 24 marbles (Priya = 14-6 = 8; Sam = 8×3 = 24).

19. C) 13 (the rule is: double the first number, then add the second).

20. 27 is the odd one out — the other four (8, 20, 24, 32) are all multiples of 4, but 27 is not.

21. 100g (sugar = 300-100 = 200g; butter = 200÷2 = 100g).

22. 16 (same rule as rows 1 and 2: double the first number, then add the second).

23. 16 (the rule is: double the first number, then add the second).

24. A, B and D are correct — Ella > Finn = Grace > Harry, so Ella is heavier than Grace (A), Finn is heavier than Harry (B), and Harry is the lightest of the four (D). C and E reverse true comparisons and are false.

25. C) 22 — the other four are all numbers whose digits add up to a multiple of 3.

26. 83 items (Shop B = 25+8 = 33; Shop C = 25×2 = 50; 33+50 = 83).

27. 13 (the rule is: subtract the smaller number from the larger).

28. C) £31 (child ticket = £12-5 = £7; total = 2×£12 + £7 = £31).

29. 33 is the odd one out — the other four (23, 29, 31, 37) are all prime numbers, but 33 is not.

30. C) 14 (the rule is: subtract the smaller number from the larger).

Standard / Advanced

31. 85 is the odd one out — the other four (63, 72, 81, 90) are all numbers whose digits add up to 9, but 85 is not.

32. 9 (the rule is: multiply the two numbers together, then subtract their sum from the result).

33. 27 cm (pencil = 5+4 = 9 cm; ribbon = 9×3 = 27 cm).

34. C) 16 (the rule is: add the two numbers together, then double the result).

35. 40 is the odd one out — the other four (21, 28, 36, 45) are all triangular numbers (the sum of the first n whole numbers), but 40 is not.

36. £200 (Year 6 = £120+40 = £160; Year 4 = £160÷2 = £80; Year 4 + Year 5 = £80+£120 = £200).

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37. 28 (same rule as rows 1 and 2: square the first number, then subtract the second).

38. 22 (the rule is: add the two numbers together, then double the result).

39. A, B, E are correct (each gives 11 using the rule: multiply the two numbers together, then subtract their sum from the result).

40. D) 200 — the other four are all perfect cubes.

41. 100 minutes (Runner Y = 185-15 = 170 min; Runner Z = 170÷2 = 85 min; 185-85 = 100).

42. 40 (the rule is: square the first number, then subtract the second).

43. C) 144 (Branch 2 = 48-12 = 36; Branch 3 = 36×3 = 108; 36+108 = 144).

44. 92 is the odd one out — the other four (48, 57, 66, 75) are all numbers whose digits add up to 12, but 92 is not.

45. C) 26 (the rule is: add the two numbers together, then double the result).

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